

Schwarzschild Tensor Calculations

Metric Tensor

$$g_{\mu\nu} = \begin{bmatrix} \frac{2M}{r} - 1 & 0 & 0 & 0 \\ 0 & -\frac{1}{\frac{2M}{r} + 1} & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2(\theta) \end{bmatrix}$$

Christoffel Symbols

$$\Gamma_{01}^0 = \frac{M}{r(-2M+r)}$$

$$\Gamma_{00}^1 = \frac{M(-2M+r)}{r^3}$$

$$\Gamma_{11}^1 = \frac{M}{r(2M-r)}$$

$$\Gamma_{22}^1 = 2M - r$$

$$\Gamma_{33}^1 = (2M - r) \sin^2(\theta)$$

$$\Gamma_{12}^2 = \frac{1}{r}$$

$$\Gamma_{33}^2 = -\frac{\sin(2\theta)}{2}$$

$$\Gamma_{13}^3 = \frac{1}{r}$$

$$\Gamma_{23}^3 = \frac{1}{\tan(\theta)}$$

Riemann Tensor

$$R_{101}^0 = \frac{2M}{r^2(-2M+r)}$$

$$R_{202}^0 = -\frac{M}{r}$$

$$R_{303}^0 = -\frac{M \sin^2(\theta)}{r}$$

$$R_{001}^1 = \frac{2M(-2M+r)}{r^4}$$

$$R_{212}^1 = -\frac{M}{r}$$

$$R_{313}^1 = -\frac{M \sin^2(\theta)}{r}$$

$$R_{002}^2 = \frac{M(2M-r)}{r^4}$$

$$R_{112}^2 = \frac{M}{r^2(-2M+r)}$$

$$R_{323}^2 = \frac{2M \sin^2(\theta)}{r}$$

$$R_{003}^3 = \frac{M(2M-r)}{r^4}$$

$$R_{113}^3 = \frac{M}{r^2(-2M+r)}$$

$$R_{223}^3 = -\frac{2M}{r}$$

Ricci Tensor

$$R_{\mu\nu} = 0$$

Ricci Scalar

$$R = 0$$

Einstein Tensor

$$G_{\mu\nu} = 0$$